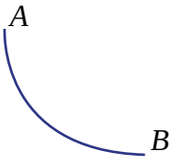



**The shortest distance path between *A* and *B***
***Fig. 11.1 (a)***




**The shortest time route between *A* and *B***
***Fig. 11.1 (b)***

Certainly this route is not the shortest time route joining the points *A* and *B*, because the velocity of the motion in the straight line (Fig. 11.1(a)) will be comparatively slow; whereas it take a curve that is steeper near *A* (Fig. 11.1(b)), even though the path becomes longer, a considerable portion of the distance will be covered at a greater speed. The solution to this problem is solved by **Integral calculus.** This is called Brachistochrone problem which initiates the study of calculus of variation using integral tool.

### Situation 2

In elementary geometry we have learnt to evaluate the measure of the following regular shape of the figures given below by using known formulae.

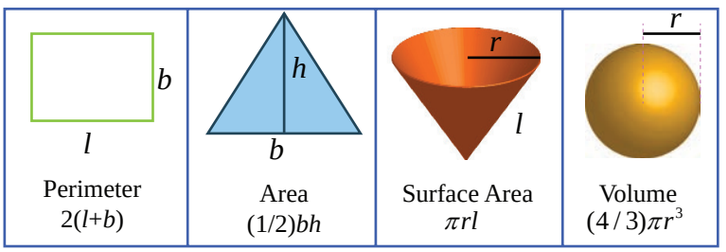



***Fig. 11.2 (a)***

How can the measure of the following figures given by functions be calculated?

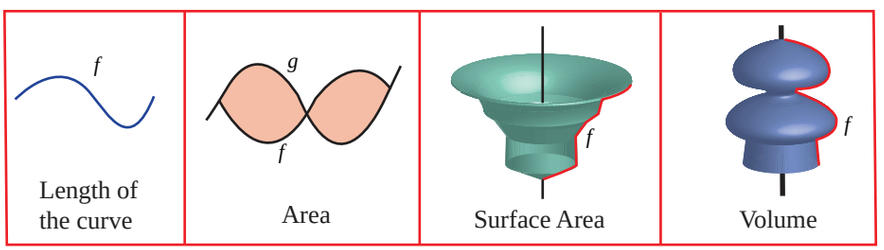



***Fig. 11.2 (b)***

Though the problems look so difficult, integral calculus solves it without any difficulties.

### Situation 3

At a particular moment, a student needs to stop his speedy bike to avoid a collision with the barrier ahead at a distance 40 metres away from him. Immediately he slows (acceleration) down the bike applying brake at a rate of 8 meter/second$^2$. If the bike is moving at a speed of 24m/s, when the brakes are applied, will it stop before collision?

Also look at the following problems that occur naturally in our life.



***Fig. 11.3***



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